

Fig. 4.46

Now in (Fig.4.46), consider the region which is surrounded by the chord PQ and major arc $\overset{\frown}{QP}$. This is called the major segment of the circle. In the same way, the segment containing the minor arc and the same chord is called the minor segment.

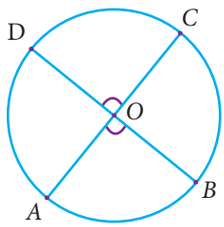


Fig. 4.47

In (Fig.4.47), if two arcs $\overset{\frown}{AB}$ and $\overset{\frown}{CD}$ of a circle subtend the same angle at the centre, they are said to be congruent arcs and we write,

$\overset{\frown}{AB} \equiv \overset{\frown}{CD}$ implies $m\overset{\frown}{AB} = m\overset{\frown}{CD}$

implies $\angle AOB = \angle COD$

Note

A diameter of a circle is:

- the line segment which bisects the circle.
- the largest chord of a circle.
- a line of symmetry for the circle.
- twice in length of a radius in a circle.

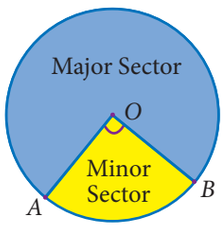


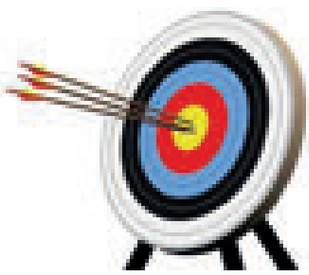
Fig. 4.48

Now, let us observe (Fig.4.48). Is there any special name for the region surrounded by two radii and arc? Yes, its name is sector. Like segment, we find that the minor arc corresponds to the minor sector and the major arc corresponds to the major sector.

Concentric Circles

Circles with the same centre but different radii are said to be concentric.

Here are some real-life examples:


An Archery target


A carom board coin


Water ripples

Fig. 4.49

Congruent Circles

Two circles are congruent if they are copies of one another or identical. That is, they have the same size. Here are some real life examples:

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